Dear interviewers:

Thank you for taking the time to review my report on question 2, "**Application for Lending Department of a Bank Part 1: Modeling of a balance sheet**". We begin with a literature review of the balance sheet modelling. Next, a point-to-point response is provided for each question. A codebase for each question is attached in the subfolder on GitHub. You can refer to more details in my github repo.

Yours sincerely,
Lixin Tu

Application for Lending Department of a Bank

Comment

Your task is to create an application that analyzes the financial health of businesses seeking loans, enabling the bank to make informed, data-driven lending decisions. This challenge requires a combination of analytical skills and innovative problem-solving. By accurately assessing risk and potential returns, your solution will contribute to the bank's ability to allocate capital efficiently and profitably.

1 Part 1. Modeling of a balance sheet

Comment 1.1

We would like to forecast the balance sheet of a company. Unfortunately, the different fields of a balance sheet are not independent. Hence we have to construct a model that respect these identities. For a short introduction of the problem, please consider the papers Pareja(07), Pareja(09) and Pelaez(11). For a much more detail exposition of the problem, please consult Shahnazarian(04) and the textbook "financial forecasting, analysis and modelling" by Samonas, as well as other standard accounting textbooks.

Pareja(07), Pareja(09) and Pelaez(11)
Shahnazarian(04)

Response:

Below is a brief literature review on balance sheet modelling.

The Balance Sheet provides a snapshot of a firm at a specific moment in time, listing its assets and rights along with the sources of financing through liabilities and equity. The objective of this project is to forecast a company's balance sheet. Conventional forecasting methods often rely on a "plug," a formulaic adjustment that forces the Balance Sheet to match. Although simple, this practice poses risks because errors in the financial statements may remain undetected as long as the plug ensures numerical consistency.

Vélez-Pareja proposed a method to construct financial statements without relying on plugs. The approach applies the double-entry principle—total assets minus total liabilities equals equity—to derive the balance sheet and forecast the financial statements in a consistent manner [Mejía-Peláez and Vélez-Pareja, 2011]. Another study introduced a simplified financial planning model that integrates the balance sheet (BS), the income statement (IS), and the cash budget (CB) [Vélez-Pareja, 2011]. They demonstrated that these statements can be constructed without plugs by developing a set of intermediate tables. Once these tables are created, the IS and BS follow directly by linking their corresponding values. The framework also incorporates the Fisher equation to compute nominal interest rates that account for inflation. Another issue is the circularity problem between firm value and cost of capital. The issue arises because the discount rate used to value cash flows depends on the value of those cash flows [Velez-Pareja, 2010]. Previous treatments either ignored the problem, assumed a constant cost of capital, eliminated taxes, or relied on manual or spreadsheet-based iterations. Building on the core financial relationships among present value, cash flow, and the discount rate, the authors derived a general, non-circular formulation for equity value, the cost of levered equity, the levered firm value, and the weighted average cost of capital. They also recommended checking any solution with methods such as Adjusted Present Value (APV) and Capital Cash Flow (CCF). Shahnazarian focused on forecasting the balance sheet with an emphasis on key components such as liquidity ratios, leverage trends, and capitalization. These indicators are essential inputs for bank credit risk models and loan portfolio stress tests. The study proposed a micro-simulation framework for firms' financial statements to model financial behaviour over time [Shahnazarian, 2004]. This approach enables the simulation of solvency and liquidity ratios under different assumptions about interest rates and economic growth, and it provides insights into the evolution of indebtedness and equity positions. Such forecasts are helpful not only for tax revenue projections but also for evaluating financial robustness.

Recent research has also explored machine learning methods for financial forecasting [Wasserbacher and Spindler, 2021]. Accurate and timely forecasts are crucial in dynamic market environments [Becker et al., 2016]. Machine learning techniques are particularly suitable for forecasting tasks because of their focus on predictive performance and their ability to identify complex patterns that generalize well beyond the training sample [Mullainathan and Spiess, 2017]. However, while machine learning excels at prediction, it is typically less equipped for parameter interpretation, which is important for counterfactual and policy analysis. One study [Gajewar and Bansal, 2016] compared random forests with traditional time-series models such as ARIMA, ETS, and STL for forecasting quarterly revenues by region and globally up to one year ahead. Their results show that a random forest with a restricted feature set outperformed both classical statistical models and internal expert forecasts at Microsoft. Another study [Barker et al., 2018] developed a machine learning–based system for quarterly revenue forecasting across thirty products and three business segments, employing methods including elastic net, random forest, K-nearest neighbors, and support vector machines. Petropoulos et al. proposed a deep learning approach for dynamic balance sheet stress testing [Petropoulos et al., 2022]. Existing stress-testing methodologies have been criticized for making unrealistic assumptions and oversimplifications. The authors introduced a holistic framework situated at the intersection of computational finance and statistical machine learning. By leveraging the expressive capabilities of deep neural networks, the approach aims to improve the predictive performance of balance sheet stress-testing models and to provide a more principled methodology for capturing the risks embedded in financial institutions' balance sheets.

Comment 1.2

Construct a very simple model of the balance sheet based on the tools of Velez-Pareja(09) and Velez-Pareja(10). Please write down the mathematical equations government the evolution of the fields of balance sheet. Is it possible to model this problem as a time series? How do we handle the accounting identities?

Response:

Mathematical Evolution of the Balance Sheet To construct a simple dynamical model, we define the state of the balance sheet at time t as a vector S_t:

$$S_t = \begin{bmatrix} A_t \\ L_t \\ E_t \end{bmatrix} \tag{1}$$

where A_t represents Total Assets, L_t represents Total Liabilities, and E_t represents Total Equity. The evolution of these fields is governed by the flow variables (changes) over the period $[t-1, t]$. The governing equations for the evolution of the system can be written as difference equations:

$$\begin{aligned} A_t - A_{t-1} + \Delta A_t \\ L_t - L_{t-1} + \Delta L_t \\ E_t - E_{t-1} + NI_t - D_t + \Delta K_t \end{aligned} \tag{2}$$

where: NI_t is Net Income (from the Income Statement). D_t represents Dividends paid. ΔK_t represents net issuance or repurchase of equity capital.

Modeling as a Time Series This problem can be considered as time-series modeling. The state S_t is strongly dependent on the previous state S_{t-1} and a vector of inputs. Therefore, we can model this as a multivariate time series system:

$$S_t = f(S_{t-1}, S_{t-2}, \ldots, S_{t-k}, X_t) + \epsilon_t \tag{3}$$

where f is the transition function (which can be linear or non-linear), X_t contains covariates, and ϵ_t is the noise term.

Handling Accounting Identities The fundamental accounting identity is a hard constraint:

$$A_t = L_t + E_t \tag{4}$$

Or equivalently in terms of flows:

$$\Delta A_t = \Delta L_t + \Delta E_t \tag{5}$$

In the next, we handle this identity using a hierarchical approach. We predict all variables (A, L, and E) independently to capture their individual time-series dynamics, and then apply a post-processing projection step to project the predicted vector onto the subspace defined by the constraint $A - L - E = 0$.

Comment 1.3

Implement the model in Tensorflow and python

Response:

We design the model in two stages:

1. Base Forecaster: An LSTM (or Dense) network that predicts Assets (A), Liabilities (L), and Equity (E) independently based on historical data.
2. Reconciliation Layer: A custom projection layer that forces the output to satisfy $A = L + E$ by distributing the "accounting error" across the three variables.

It should be noted that this implementation assumes "equal variance" (it adjusts A, L, and E equally).

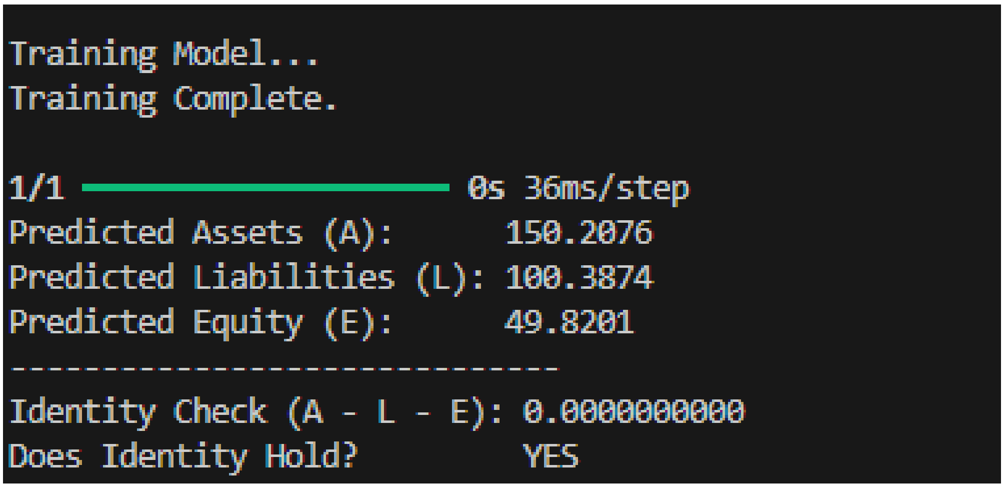

Figure 1: A screenshot of the basic model implementation and identity verification.

Comment 1.4

You can get income statement and balance sheet data from yahoo finance. This blog post may help you. https://rfachrizal.medium.com/how-to-obtain-financial-statements-from-stocks-using-yfinance-87c432b803b8

Response:

Herein, we followed the instructions to retrieve income statement and balance sheet data for several companies, including Tencent, Alibaba, JPMorgan Chase, ExxonMobil, Volkswagen, Microsoft, and Google. The data were collected from yfinance and cover the past four years.

Table 1: Balance Sheet-style presence matrix (Tencent, JPMorgan, Alibaba_HK, Exxon, Volkswagen, Microsoft, Google)

Item (from yahoo finance)	**Tencent**	**JPMorgan**	**Alibaba_HK**	**Exxon**	**VW**	**Microsoft**	**Google**
ASSETS							
Current Assets							
Current Assets	✓		✓	✓	✓	✓	✓
Cash And Cash Equivalents	✓	✓	✓	✓	✓	✓	✓
Cash Equivalents	✓					✓	

(Continued on next page)

(Continued from previous page)

Item (BS grouping)	Tencent	JPMorgan	Alibaba_HK	Exxon	VW	Microsoft	Google
Cash Financial	✓	✓			✓	✓	
Cash Cash Equivalents And Short Term Investments	✓		✓	✓	✓	✓	✓
Other Short Term Investments	✓	✓	✓		✓	✓	✓
Receivables		✓	✓	✓		✓	✓
Accounts Receivable	✓	✓	✓	✓	✓	✓	✓
Gross Accounts Receivable	✓			✓		✓	✓
Allowance For Doubtful Accounts Receivable	✓			✓		✓	✓
Other Receivables	✓	✓	✓	✓	✓		
Accrued Interest Receivable			✓				
Taxes Receivable	✓		✓		✓		✓
Inventory	✓		✓	✓	✓	✓	✓
Finished Goods				✓	✓	✓	
Work In Process					✓	✓	
Raw Materials				✓	✓	✓	
Other Inventories					✓		
Prepaid Assets	✓		✓		✓		
Other Current Assets	✓		✓	✓	✓	✓	✓
Hedging Assets Current	✓				✓	✓	
Assets Held For Sale Current					✓		
Restricted Cash	✓		✓	✓	✓		
Current Deferred Assets			✓				
Duefrom Related Parties Current			✓				
Non-current Assets							
Total Non Current Assets	✓		✓	✓	✓	✓	✓
Net PPE	✓	✓	✓	✓	✓	✓	✓
Gross PPE	✓	✓	✓	✓	✓	✓	✓
Accumulated Depreciation	✓		✓	✓	✓	✓	✓
Construction In Progress	✓		✓		✓		✓
Leases						✓	✓
Machinery Furniture Equipment	✓		✓		✓	✓	✓
Buildings And Improvements	✓		✓		✓	✓	✓
Land And Improvements						✓	✓
Other Properties	✓	✓	✓	✓	✓	✓	✓
Properties	✓	✓			✓		✓
Goodwill And Other Intangible Assets	✓	✓	✓		✓	✓	✓
Goodwill	✓	✓	✓		✓	✓	✓
Other Intangible Assets	✓	✓	✓		✓	✓	✓
Investments And Advances		✓	✓	✓		✓	✓
Investmentin Financial Assets	✓		✓	✓	✓	✓	✓
Available For Sale Securities	✓	✓	✓	✓	✓	✓	✓
Held To Maturity Securities		✓					
Trading Securities		✓					
Financial Assets Designatedas Fair Value Through Profitor Loss Total	✓						
Long Term Equity Investment	✓		✓	✓	✓	✓	
Investmentsin Joint Venturesat Cost	✓						
Investmentsin Associatesat Cost	✓						
Investment Properties	✓				✓		
Financial Assets	✓				✓	✓	
Other Non Current Assets	✓		✓	✓	✓	✓	✓
Non Current Prepaid Assets	✓			✓			

(Continued on next page)

(Continued from previous page)

Item (BS grouping)	Tencent	JPMorgan	Alibaba_HK	Exxon	VW	Microsoft	Google
Non Current Deferred Taxes Assets	✓		✓		✓		✓
Non Current Deferred Assets			✓				✓
Non Current Accounts Receivable				✓			
Totals							
Total Assets	✓	✓	✓	✓	✓	✓	✓
LIABILITIES							
Current Liabilities							
Current Liabilities	✓		✓	✓	✓	✓	✓
Current Debt And Capital Lease Obligation	✓	✓	✓	✓	✓	✓	✓
Current Debt	✓	✓	✓	✓	✓	✓	
Current Capital Lease Obligation	✓		✓		✓	✓	✓
Other Current Borrowings		✓	✓	✓		✓	
Line Of Credit				✓			
Commercial Paper		✓		✓		✓	
Payables And Accrued Expenses		✓	✓	✓	✓	✓	✓
Payables	✓	✓	✓	✓	✓	✓	✓
Accounts Payable	✓	✓	✓	✓	✓	✓	✓
Other Payable	✓	✓	✓	✓	✓		✓
Current Accrued Expenses			✓				✓
Total Tax Payable	✓		✓	✓	✓	✓	✓
Income Tax Payable			✓	✓		✓	✓
Other Current Liabilities			✓		✓	✓	✓
Current Deferred Liabilities			✓			✓	✓
Current Deferred Revenue			✓			✓	✓
Pensionand Other Post Retirement Benefit Plans Current	✓				✓	✓	✓
Current Provisions					✓		
Dueto Related Parties Current			✓	✓			
Non-current Liabilities							
Total Non Current Liabilities Net Minority Interest	✓		✓	✓	✓	✓	✓
Long Term Debt And Capital Lease Obligation	✓	✓	✓	✓	✓	✓	✓
Long Term Debt	✓	✓	✓	✓	✓	✓	✓
Long Term Capital Lease Obligation	✓		✓	✓	✓	✓	✓
Capital Lease Obligations	✓		✓	✓		✓	✓
Non Current Deferred Revenue	✓		✓		✓	✓	✓
Non Current Deferred Taxes Liabilities	✓	✓	✓	✓	✓	✓	✓
Tradeand Other Payables Non Current	✓				✓	✓	✓
Other Non Current Liabilities			✓	✓	✓	✓	✓
Non Current Deferred Liabilities			✓	✓		✓	✓
Non Current Pension And Other Postretirement Benefit Plans				✓	✓		
Employee Benefits				✓			
Non Current Accrued Expenses					✓		
Derivative Product Liabilities					✓		
Long Term Provisions			✓		✓		
Dueto Related Parties Non Current				✓			
Totals / Funding							
Total Liabilities Net Minority Interest	✓	✓	✓	✓	✓	✓	✓
Total Debt	✓	✓	✓	✓	✓	✓	✓

(Continued on next page)

(Continued from previous page)

Item (BS grouping)	**Tencent**	**JPMorgan**	**Alibaba_HK**	**Exxon**	**VW**	**Microsoft**	**Google**
EQUITY							
Total Equity Gross Minority Interest	✓	✓	✓	✓	✓	✓	✓
Stockholders Equity	✓	✓	✓	✓	✓	✓	✓
Common Stock Equity	✓	✓	✓	✓	✓	✓	✓
Capital Stock	✓	✓	✓	✓	✓	✓	✓
Common Stock	✓	✓	✓	✓	✓	✓	✓
Preferred Stock		✓					✓
Additional Paid In Capital	✓	✓	✓		✓		
Retained Earnings	✓	✓	✓	✓	✓	✓	✓
Treasury Stock	✓	✓	✓	✓			
Gains Losses Not Affecting Retained Earnings		✓	✓	✓		✓	✓
Other Equity Adjustments		✓	✓	✓		✓	✓
Foreign Currency Translation Adjustments			✓				
Unrealized Gain Loss			✓				
Other Equity Interest	✓				✓		
Preferred Stock Equity		✓					
Minority Interest	✓		✓	✓	✓		
Ordinary Shares Number	✓	✓	✓	✓	✓	✓	✓
Preferred Shares Number		✓					
Treasury Shares Number	✓	✓	✓	✓			✓
MEMO / DERIVED							
Working Capital	✓		✓	✓	✓	✓	✓
Tangible Book Value	✓	✓	✓	✓	✓	✓	✓
Net Tangible Assets	✓	✓	✓	✓	✓	✓	✓
Invested Capital	✓	✓	✓	✓	✓	✓	✓
Total Capitalization	✓	✓	✓	✓	✓	✓	✓
Net Debt	✓		✓	✓	✓	✓	
Cash Cash Equivalents And Federal Funds Sold		✓					

Based on the results in Table 1, we construct a three–level hierarchy of balance sheet items available to all seven firms.

Level 0: All–firms aggregates.

- All-firms *Total Assets*,
- All-firms *Total Liabilities Net Minority Interest*,
- All-firms *Total Equity Gross Minority Interest*,

Level 1: Firm-level totals.

$$\mathcal{F} = \{\text{Tencent}, \text{JPMorgan}, \text{Alibaba\_HK}, \text{Exxon}, \text{Volkswagen}, \text{Microsoft}, \text{Google}\}.$$

For each firm $i \in \mathcal{F}$, we include the following total series:

- $\textit{Total Assets}_i$,
- $\textit{Total Liabilities Net Minority Interest}_i$,
- $\textit{Total Equity Gross Minority Interest}_i$,
- $\textit{Invested Capital}_i$.

Level 2: Common balance-sheet components within each firm.

Assets branch. For firm i, the asset side is decomposed as

- *Cash And Cash Equivalents*,
- *Accounts Receivable*,
- *Net PPE*,
- *Available For Sale Securities*,
- *Other assets*

Liabilities branch. For firm i, the liability side is decomposed as

- *Current Debt And Capital Lease Obligation*,
- *Long Term Debt And Capital Lease Obligation*,
- *Payables*, with *Accounts Payable* as the core component,
- *Non Current Deferred Taxes Liabilities*,
- *Other liabilities*

Equity branch. For firm i, the equity side is decomposed as

- *Capital Stock*,
- *Common Stock*,
- *Retained Earnings*,
- *Other equity*

Comment 1.5

Choose some companies to apply your model to. How are you going to train your model? How can you test if your model is good at forecasting the balance sheet of the company? How can you ensure that your forecast at least respect the accounting identities, and at least satisfying the asset = liability + equity identity as other relationship stated in the papers quoted here?

Response:

Modeling Strategy for Short Annual Series Given that the above annual financial reports provide short time series (T=5 years), deep learning approaches (like LSTM/transformer) are prone to overfitting. To address this, we adopt a Global Hierarchical Forecasting approach (also known as a Panel Data approach). Instead of training separate models for each firm, we train a single, shared mapping function across all firms simultaneously. The model takes the balance sheet state at year t as input and predicts the state at year $t+1$. By pooling data from all firms in the set $\mathcal{F}$, we leverage the cross-sectional commonalities in financial accounting principles, effectively increasing our training sample size relative to the model's parameter count.

Handling Accounting Identities Despite the short time horizon, preserving accounting identities remains crucial. We employ a Constrained Geometric Reconciliation layer implemented in TensorFlow. A parsimonious neural network (Global MLP) predicts the bottom-level components (Level 2) based on the previous year's values. These component predictions are then projected through a fixed Summation Matrix (S) to derive firm-level aggregates (Level 1) and global aggregates (Level 0). This guarantees that $Total = \sum Components$ holds exactly by design, ensuring internal consistency.

Evaluation We adopt a Leave-One-Year-Out validation scheme suitable for short series. We train the global model on transitions from Year 1 to Year $T - 1$ and evaluate its ability to predict the Year T balance sheet (the final year). Performance is measured by the discrepancy in accounting identities and the Mean Absolute Error (MAE) of the forecasted values.

The code implementation is based on the synthetic data for simplicity. However, all trained indices are derived from the above analysis.

Comment 1.6

Can you use your model to forecast earnings?

Response:

Yes, our model can be modified to infer earnings forecasts, specifically through the Clean Surplus Relation (CSR). While our primary architecture targets the Balance Sheet (stock variables), the Income Statement (flow variables) is linked to the Balance Sheet through the Retained Earnings account. The evolution of Retained Earnings (RE_t) is governed by:

$$RE_t = RE_{t-1} + NI_t - Div_t \tag{6}$$

where:NI_t is the Net Income (Earnings) for period $t.Div_t$ represents Dividends paid in period t.

Since our Global HTS model forecasts the future state of all balance sheet components, it predicts $\widehat{RE}_t$. By rearranging the equation, we can derive the Implied Net Income ($\widehat{NI}_t$):

$$\widehat{NI}_t = (\widehat{RE}_t - RE_{t-1}) + \widehat{Div}_t \tag{7}$$

In our implementation, we assume a constant dividend payout ratio or a martingale process where $Div_t \approx Div_{t-1}$. Thus, the model's forecast of structural equity growth provides a direct estimate of the firm's profitability.

Comment 1.7

What are the ML techniques we can use to your model to make it better?

Response:

Our current Global MLP architecture is chosen for resolving the data scarcity issue. We also identify three specific ML directions that could significantly improve the model's performance and robustness.

Modeling Firm Dependencies with Graph Neural Networks (GNNs) Our current model treats the set of firms $\mathcal{F}$ as a pooled cross-section, assuming independence between firms. However, financial entities are interconnected through supply chains and sector similarities. To capture this, we can model the dataset as a graph with nodes representing firms and edges representing relationships [Chen et al., 2018]. Graph Convolutional Networks (GCNs) or Relation-aware GNNs can propagate information across these edges, allowing the balance sheet forecast of one firm (e.g., Tencent) to be informed by the state of its peers (e.g., Alibaba).

Enhancing Interpretability with Temporal Fusion Transformers (TFT) If longer time series become available, we can also transition from simple dense layers to Temporal Fusion Transformers (TFT). Unlike standard LSTMs, TFTs employ "Variable Selection Networks" and multi-head attention mechanisms [Lim et al., 2021]. This would allow the model to automatically learn which balance sheet components (e.g., Inventory vs. Receivables) are most predictive of future states.

Hard Constraints via Differentiable Optimization Layers In our current implementation, we use a custom projection layer to enforce $A = L + E$. To generalize this, we can incorporate OptNet (Differentiable Optimization Layers). Instead of a simple algebraic projection, OptNet treats the reconciliation step as a Quadratic Programming (QP) problem embedded inside the neural network [Amos and Kolter, 2017]. This allows learning complex inequality constraints (e.g., $Cash \geq 0$, $Debt \geq 0$) alongside equality constraints, while maintaining full end-to-end differentiability.

Comment 1.8

Hint: simulation is highly related to prediction. Suppose that you can simulate $y(t+1)$ given $y(t)$. The prediction problem is very simple to implement numerically. A general form of the model can be written as $y(t+1) = f(x(t), y(t)) + n(t)$, where $n(t)$ is some noise term to be specified, and $x(t)$ are additional sets of variables that are relevant for the simulation. What should $x(t)$ be?

Response:

Following the general form

$$y(t+1) = f(x(t), y(t)) + n(t), \tag{8}$$

where $y(t)$ is the vector of balance sheet states. We can define $x(t)$ as the vector of Exogenous Drivers and Control Variables. Specifically, in the context of financial statement forecasting, $x(t)$ should contain the external and operational forces that "drive" the balance sheet from one state to the next. We categorize $x(t)$ into key components as follows.

1. **Operational Drivers (Flow Variables)**: The most critical element in $x(t)$ is Revenue (Sales) or Revenue Growth. Revenue is the primary driver of working capital. For example, Accounts Receivable at $t+1$ is functionally a function of Accounts Receivable at t and Revenue at t (via the Days Sales Outstanding metric). Similarly, Inventory is driven by Cost of Goods Sold (COGS).

2. **Macroeconomic State Variables**: These are external economic conditions that influence valuation and debt obligations.Interest Rates (r_t): These drive the evolution of Financial Liabilities (via interest expense and debt service capacity). GDP Growth / Sector Indices: These serve as systemic indicators that correlate with the firm's asset expansion.

This is the end of Part 1. We would appreciate any suggestions you may have.
In Part 2, we will apply a large language model to financial statement analysis.

References

[Amos and Kolter, 2017] Amos, B. and Kolter, J. Z. (2017). Optnet: Differentiable optimization as a layer in neural networks. In *International conference on machine learning*, pages 136–145. PMLR.

[Barker et al., 2018] Barker, J., Gajewar, A., Golyaev, K., Bansal, G., and Conners, M. (2018). Secure and automated enterprise revenue forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32.

[Becker et al., 2016] Becker, S. D., Mahlendorf, M. D., Schäffer, U., and Thaten, M. (2016). Budgeting in times of economic crisis. *Contemporary Accounting Research*, 33(4):1489–1517.

[Chen et al., 2018] Chen, Y., Wei, Z., and Huang, X. (2018). Incorporating corporation relationship via graph convolutional neural networks for stock price prediction. In *Proceedings of the 27th ACM international conference on information and knowledge management*, pages 1655–1658.

[Gajewar and Bansal, 2016] Gajewar, A. and Bansal, G. (2016). Revenue forecasting for enterprise products. *arXiv preprint arXiv:1701.06624*.

[Lim et al., 2021] Lim, B., Arık, S. Ö., Loeff, N., and Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International journal of forecasting*, 37(4):1748–1764.

[Mejía-Peláez and Vélez-Pareja, 2011] Mejía-Peláez, F. and Vélez-Pareja, I. (2011). Analytical solution to the circularity problem in the discounted cash flow valuation framework. *Innovar*, 21(42):55–68.

[Mullainathan and Spiess, 2017] Mullainathan, S. and Spiess, J. (2017). Machine learning: An applied econometric approach. *Journal of Economic Perspectives*, 31(2):87–106.

[Petropoulos et al., 2022] Petropoulos, A., Siakoulis, V., Panousis, K. P., Papadoulas, L., and Chatzis, S. (2022). A deep learning approach for dynamic balance sheet stress testing. In *Proceedings of the Third ACM International Conference on AI in Finance*, ICAIF '22, page 53–61. ACM.

[Shahnazarian, 2004] Shahnazarian, H. (2004). *A Dynamic Microeconometric Simulation Model for Incorporated Businesses*. Sveriges riksbank.

[Velez-Pareja, 2010] Velez-Pareja, I. (2010). Constructing consistent financial planning models for valuation. *IIMS Journal of Management of Science*, 1.

[Vélez-Pareja, 2011] Vélez-Pareja, I. (2011). Forecasting financial statements with no plugs and no circularity. *The IUP Journal of Accounting Research & Audit Practices*, 10(1).

[Wasserbacher and Spindler, 2021] Wasserbacher, H. and Spindler, M. (2021). Machine learning for financial forecasting, planning and analysis: recent developments and pitfalls. *Digital Finance*, 4(1):63–88.